

CHAPTER 1

Introduction

1.1 Overview

In the contemporary landscape of industrial innovation and technological advancement, the integration of smart solutions has become increasingly pivotal across various sectors. One such domain that has witnessed transformative developments is the mining industry, a cornerstone of economic prosperity for nations worldwide. This paper delves into the critical intersection of technology and mining safety, addressing the multifaceted challenges faced by miners and proposing an innovative approach to enhance their well-being and security. The benefits generated by this industry contribute significantly to local communities by processing the materials it offers. However, working in mining poses specific health and safety risks, especially in challenging or unpredictable conditions. As mines expand, the risks associated with completing tasks also increase. The mining industry is complex, involving intricate operations conducted within tunnels, underground passages, and other challenging environments. The intricate nature of mining operations presents a variety of risk variables that may compromise the well-being and security of miners. The Chasnala mining tragedy in the Indian state of Jharkhand, close to Dhanbad, is a heartbreaking example. Almost 372 miners' lives were almost lost in this tragedy, which is regarded as one of the deadliest in the history of the mining industry.

Miners frequently encounter unnoticed environmental factors such as changes in pressure and temperature. The lives of excavators are in grave danger when they crash with big things like hard rocks or mining equipment. The inhalation of dangerous gases poses a substantial risk of injury to miners and is considered a severe threat. Miners are cut off from outside communication under such circumstances.

Recognizing the need for proactive safety measures, this paper explores the development of a smart protective helmet system designed to detect and respond to hazardous events in real-time. Beyond event detection, the system encompasses environmental monitoring, GPS tracking, and the provision of oxygen enhancements to mitigate risks associated with toxic gases. The proposed system not only addresses the immediate safety concerns but also serves as a forward-looking initiative to ensure the well-being of miners.

Moreover, considering the rising prominence of the mining sector in certain regions, such as Pakistan with its substantial coal reserves, the paper discusses the challenges faced in making sure that miners are safe. The convergence of technological solutions, such as microcontroller-based monitoring systems and the Internet of Things (IoT), presents as a pivotal strategy to overcome these challenges and fortify the safety infrastructure within the mining industry. In navigating this discourse, the aim is to contribute to the ongoing dialogue on mining safety by presenting an integrated and forward-thinking approach. By leveraging cutting-edge technologies and innovative solutions, we strive to redefine the safety paradigm for miners, fostering a safer, more secure environment for those working at the heart of economic prosperity.

1.1 Objectives

- ✓ To develop an advanced smart helmet system with a diverse set of sensors, including environmental sensors (such as gas, temperature, and humidity), an Accelerometer and Gyroscope sensor for fall detection, IR sensor for helmet usage verification and Heart Rate sensor to measure the heart pulse rate.
- ✓ To Implement an intelligent alert system using voice notifications through the APR33A3 voice playback module.

1.2 Problem Statement

In underground mining there is a concern about the safety of the workers due to its highly changing environment. Thousands of miners die from mining accidents every year.

To save the workers life and to improve safety in mining environment it is important to take some safety measures and to improve communication between workers and control stations to avoid life threatening situations.

With the help of the smart helmet, we can provide security and rescue measures in case of any emergency conditions.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey

[1] Title This paper introduces an Intelligent Helmet system equipped with various sensors and utilizing Zigbee protocol for real-time monitoring of hazardous conditions. The proposed system integrates multiple sensors, including temperature, methane gas, and heart rate, with a Zigbee mesh network ensuring reliable data transmission for timely alerting and emergency response.

[2] This System proposed a wearable IoT-enabled jacket specifically crafted to safeguard individuals employed in coal mines, often subjected to potential hazards. This prototype is engineered to detect multiple factors such as harmful chemicals, the heartbeat of a coal miner, underground conditions, and the miner's GPS location. The collected data is intended to be transmitted accessing an ever-changing internet protocol using an encrypted Wi-Fi channel

[3] Hazardous environments at the workplace are a significant contributor to injuries due to accidents as well as chronic diseases. There are many occupational health and safety (OHS) systems, but they are costly or not flexible. This paper presents a low-cost OHS. It consists of various sensors that can be used to monitor whether a safety helmet is being worn, the worker is mobile and safety boots are being worn. The system interfaces with a wireless sensor network and is suitable for operation in environments such as underground mines and sawmills.

[4] This paper presents implementation of safety helmet for coal mine workers. This helmet is equipped with methane and carbon monoxide gas sensor. This sensor senses the gas and data is transmitted to the control room wirelessly, through a wireless module called X-Bee connected with the helmet. When the methane or carbon-monoxide gas concentration is beyond the critical level, controller in the control room triggers.

[5] In order to solve such problems of the mine cable gas monitoring system as high costs, inconvenient maintenance, hindering wiring and unmovable monitoring information points, a mine safety helmet is designed, based on wireless sensors networks, which studies the hardware design, gas concentration calculations and its dynamic compensation algorithm, as well as the experimental design of gas monitoring system in mine safety helmet so as to ensure the correct calculation of the gas concentration. The innovative point of this essay is the combination of the ordinary LED cap lamp and the wireless sensors networks, which ensures a real-time, dynamic collection of the information of the gas concentration around the workers.

[6] Temperature and Pressure sensors are used for the continuous monitoring of environmental conditions. The information are sent to the control room through wireless network. The layout of the visualization was completed and displayed in the control room with the help of a Lab VIEW software. This paper presents the undertaken design detailing solutions to issues raised in previous research.

[7] The critical levels of the hazardous gases such as CO, SO₂ and NO₂ in the mines industry has been indicated through alerting unit. The helmet removal test was done successfully with an off the-shelf IR distance sensor. The IR sensor designed from first principles was working device.

CHAPTER 3

PLANNING

3.1 Existing System

In the existing mining system, the conventional helmet functions as a protective measure against potential hazards, ensuring the safety of miners. However, this system has inherent limitations. While the primary purpose of the helmet is to shield the head from injuries, challenges arise in maintaining environmental awareness. The weight and discomfort of the helmet often prompt miners to remove it, subjecting them to unsafe conditions. Additionally, the existing system lacks fall detection capabilities, presenting a potential risk as workers may faint and fall unexpectedly, particularly when exposed to dangerous gases in the mining area. Miners also face the danger of colliding with substantial objects like mining equipment or hard rocks, exposing them to severe life-threatening risks. In such scenarios, a fall detection sensor plays a crucial role, addressing a significant gap in the capabilities of existing helmets.

The current alert system relies on a buzzer, which activates in the event of any abnormal condition. However, it lacks specificity regarding the particular condition, making it challenging to discern the reason for the alert. Having a system that specifies the exact cause during alerts would enhance our understanding and enable us to respond more cautiously to the identified cause.

3.1.1 Disadvantages

In Traditional helmets focus primarily on protecting the head but lack sensors for real-time monitoring of environmental conditions. Miners may be unaware of changes in temperature, humidity, or the presence of hazardous gases, exposing them to potential risks.

Conventional helmets lack advanced fall detection capabilities, making it challenging to promptly identify and respond to miners who may experience unexpected falls. This limitation poses a significant risk, especially in dynamic and challenging mining environments.

The alert systems in traditional helmets typically rely on generic buzzers that activate during abnormal conditions. However, these alerts lack specificity, providing little information about the nature of the emergency. Miners may struggle to understand the exact cause, hindering their ability to respond effectively.

- The existing systems often lack efficient communication channels, especially during emergencies. Miners may face difficulties in quickly and accurately communicating their situation or requesting assistance, leading to potential delays in response times.
- Conventional helmets do not incorporate advanced sensors to detect specific environmental hazards, such as the presence of toxic gases. This limitation can result in delayed responses to dangerous conditions, putting miners at risk of exposure to harmful substances.

3.2 Proposed System

The innovation at hand represents a groundbreaking development in workplace safety – an advanced protective helmet equipped with a sophisticated array of sensors designed for exhaustive detection and analysis. This pioneering system integrates two primary categories of sensors: environmental sensors for monitoring workplace conditions and sensors dedicated to tracking the well-being of workers. Within the suite of environmental sensors, a gas sensor is employed to detect hazardous gases, while temperature and humidity sensors identify abnormal fluctuations in the work environment.

Simultaneously, the system monitors the condition of workers through the integration of specialized sensors, including a pulse sensor for real-time tracking of heart rates, an infrared sensor ensuring continuous helmet usage, and a MEMS sensor designed to detect sudden falls. The continuous monitoring by these sensors ensures a holistic approach to workplace safety.

In the event of abnormal conditions where tracked values surpass predefined thresholds, the system activates a unique sound alert. Diverging from conventional buzzers, the system employs an APR33A3 voice module, providing a detailed sound alert that specifies the exact cause of the triggered response. This precise communication is crucial for enabling swift and informed reactions to potential hazards.

Beyond localized alerts, the system ensures wider communication by sending alert messages to registered mobile numbers, accompanied by GPS location details facilitated by GSM and GPS modules. This integration enhances emergency response capabilities, allowing for timely and accurate intervention in critical situations.

To facilitate comprehensive data management, the system incorporates a WiFi module that stores information on Thing Speak. This not only ensures real-time data access but also enables data retention for future references and predictive analyses. In essence, the advanced helmet system emerges as a cutting-edge solution that goes beyond traditional safety measures. It elevates workplace safety through its capabilities for real-time monitoring, precise alerts, and the generation of data-driven insights, marking a significant leap forward in ensuring the well-being of workers in diverse industrial settings.

3.2.1 Advantages

- Virtual The proposed system integrates environmental sensors, including gas, temperature, and humidity sensors, providing comprehensive monitoring of workplace conditions. This ensures early detection of hazardous gases and abnormal fluctuations, enhancing overall safety.
- Specialized sensors such as the pulse sensor enable real-time tracking of workers' heart rates, offering continuous health monitoring. This feature enhances the system's capability to address potential health risks promptly.
- The inclusion of a MEMS sensor enables the detection of sudden falls, a crucial safety feature in environments where workers may face the risk of fainting or unexpected accidents. This significantly reduces the response time to critical situations.
- Unlike traditional alert systems, the proposed system employs an APR33A3 voice module, providing detailed and specific alerts. This enhances communication during emergencies, allowing for a more informed and timely response to identified causes.

3.3 Hardware Requirements

- [1] Microcontroller ESP32
- [2] DHT 11 Sensor
- [3] MQ-2 Gas Sensor
- [4] PCB
- [5] Jumper Wires
- [6] ☐ elmet
- [7] MPU6050
- [8] LCD Display
- [9] ESP8266 Wi-Fi Module
- [10] GPS Module
- [11] Battery

3.4 Non Functional Requirements

Microcontroller ESP32

The The Arduino is an open-source microcontroller board based on the microcontroller, providing a versatile and user-friendly platform for a wide range of applications. With 54 digital input/output pins, including 16 analog inputs and 14 PWM outputs, this board is well-equipped for various projects. It features four hardware serial ports (UARTs), a 16 MHz crystal oscillator, an ICSP header, a power jack, a USB connection, and an RST button.

Powering the Arduino is straightforward, as it can be connected to a PC via USB, powered by a battery, or connected to an AC-DC adapter. The board's design includes measures to protect it from unexpected electrical discharges, making it suitable for diverse environments. Notably, the board's adaptability extends to working with existing shields and accommodating newer shields utilizing additional pins.

The specifications of the Arduino include an operating voltage of 5 volts, a recommended input voltage ranging from 7 to 12 volts, and 54 digital I/O pins, 16 analog input pins, and 15 PWM outputs. It features a clock speed of 16 MHz, with flash memory of 256 KB (8 KB used by the bootloader), 8 KB of static random access memory (SRAM), and 4 KB of electrically erasable programmable read-only memory (EEPROM).



DHT 11 Sensor

It's fairly simple to use but requires careful timing to grab data. The only real down-side of this sensor is you can only get new data from it once every 2 seconds, so when using our library, sensor readings can be up to 2 seconds old.



Fig 3.2: DHT 11 Sensor

Compared to the DHT22, this sensor is less precise, less accurate, and works in a smaller range of temperature/humidity, but it's smaller and less expensive. The DHT11 is a commonly used Temperature and humidity sensor that comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory calibrated and hence easy to interface with other microcontrollers. DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them.

MQ-2 Gas Sensor



Sensors are the electronic devices used for interaction with the outer environment. There are various types of sensors available that can detect light, noise, smoke, proximity etc. With the advent in technology, these are available as both analog and digital forms. Besides forming a communication with the outer environment, sensors are also a crucial part of safety systems. Fire sensors are used to detect the fire and take appropriate pre-cautions on time. For smooth functioning of control systems and sensitive electronics, humidity sensors are used for maintaining humidity in the unit. One of such sensors used in safety systems to detect harmful gases is MQ2 Gas sensor.

MPU6050 Sensor (MEMS)

MEMS (Micro-Electro-Mechanical Systems) sensors represent a groundbreaking technology that seamlessly integrates miniature mechanical and electrical components on a microscopic scale. These sensors, characterized by their compact size and efficiency, encompass various types, including accelerometers, gyroscopes, magnetometers, pressure sensors, and Inertial Measurement Units (IMUs). Operating on principles like capacitance changes and piezoelectric effects, MEMS sensors detect, measure, and respond to environmental stimuli. Widely adopted across diverse industries, these sensors find applications in consumer electronics, automotive systems, healthcare devices, industrial machinery monitoring, and environmental monitoring through weather stations and IoT devices. The miniaturization of components, cost-effectiveness due to mass production, and low power consumption contribute to the widespread utilization of MEMS sensors.



Fig 3.7: MPU6050 Sensor

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In the context of fall detection, MEMS accelerometers are utilized to monitor an individual's movements. When a person falls, the accelerometer detects a sudden change in acceleration, as the device experiences a rapid downward motion. The sensor then sends this data to a processing unit, which analyzes the acceleration pattern to determine whether it corresponds to a fall event. Fall detection algorithms are programmed to differentiate between regular activities and an actual fall. Factors such as the speed, angle, and impact force are considered. If the algorithm identifies a fall, it triggers an alert or alarm, notifying caregivers, emergency services, or relevant personnel about the potential fall.

ESP8266 Wi-Fi Module

ESP8266 was designed by the Chinese company Express if Systems for uses in Internet of Things (IoT) systems. ESP8266 is a complete Wi-Fi system on chip that incorporates a 32-bit processor, some RAM and depending on the vendor between 512KB and 4MB of flash memory. This allows the chip to either function as a wireless adapter that can extend other systems with Wi-Fi functionality, or as a standalone unit that can by itself execute simple applications. Depending on the specific module variant (ESP-1 to ESP-12 at the time of this thesis) between 0 and 7 General Purpose Input/Output (GPIO) pins are available, in addition to Rx and Tx pins of the UART, making the module very suitable for IoT applications. The Software Development Kit (SDK) provided by Espresso if contains a lightweight implementation of a TCP/IP control stack (lwIP) for Wi-Fi communication. The modules house libraries for optional services such as Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), JavaScript Object Notation (JSON) and Secure Socket Layer (SSL) libraries for Application-Level programming. It incorporates 802.11 MAC extensions such as 802.11b/g/n/d/e/h/i/k/r that manage signal transmission, encapsulation, encryption, collision management and roaming functionality. The chip generally comes as part of a module, soldered to a Printed Circuit Board (PCB), however it is possible to purchase only the chip itself in order to create a truly custom module. The module variants currently available on the market may include an antenna (PCB or ceramic) or a U-FL connector, a hardware component for serial communication and a myriad of other auxiliary components such as resistors, capacitors and LEDs.

GPS Module

The GPS (Global Positioning System) module serves as a cornerstone in our smart helmet system, offering precise location tracking capabilities by interfacing with a network of orbiting satellites. This integral component facilitates real-time determination of accurate geographical coordinates, encompassing latitude, longitude, and sometimes altitude. By continuously receiving signals from multiple satellites, the GPS module ensures dynamic and real-time tracking, providing invaluable insights into the device or user's movement.



Fig 3.12: GPS Module

An essential feature of GPS modules is their ability to not only pinpoint location coordinates but also furnish additional data such as speed and direction, contributing to enhanced navigation capabilities. These modules seamlessly integrate with microcontrollers or processing units, allowing devices to incorporate location-based functionalities into their applications. The versatility of GPS technology is evident in its widespread use across diverse applications, spanning navigation systems in vehicles and smartphones to tracking devices, asset management, and outdoor recreational activities.

In our smart helmet system, the GPS module assumes a pivotal role by enabling the helmet to transmit accurate location details. This functionality proves crucial in emergency situations, as it ensures that alerts sent to registered mobile numbers include vital GPS location information. The integration of GPS technology underscores our commitment to enhancing workplace safety through real-time tracking and effective communication, positioning our system at the forefront of intelligent and responsive safety solutions.

CHAPTER 4

SYSTEM DESIGNS

4.1 Arduino IDE

The Arduino IDE is the software tool used to write, compile, and upload Arduino sketches to an Arduino board. It is a free and open-source software tool that can be downloaded from the Arduino website for Windows, Mac OS X, and Linux operating systems. The Arduino IDE provides a user-friendly interface for creating, editing, and managing Arduino sketches. It includes a text editor with features such as syntax highlighting, auto-indentation, and code completion to make programming easier and faster.

An Arduino sketch from the fig.3.16 is typically written in C or C++ programming languages, and consists of two main functions: `setup()` and `loop()`. The `setup()` function is called once when the board is powered on or reset, and is used to initialize the board and any peripherals connected to it. The `loop()` function is then called repeatedly until the board is turned off or reset, and is used to perform the main tasks of the program.

The Arduino IDE also includes a serial monitor that allows the user to view and send data to the board over a serial connection. This is useful for debugging and communicating with the board during program execution.

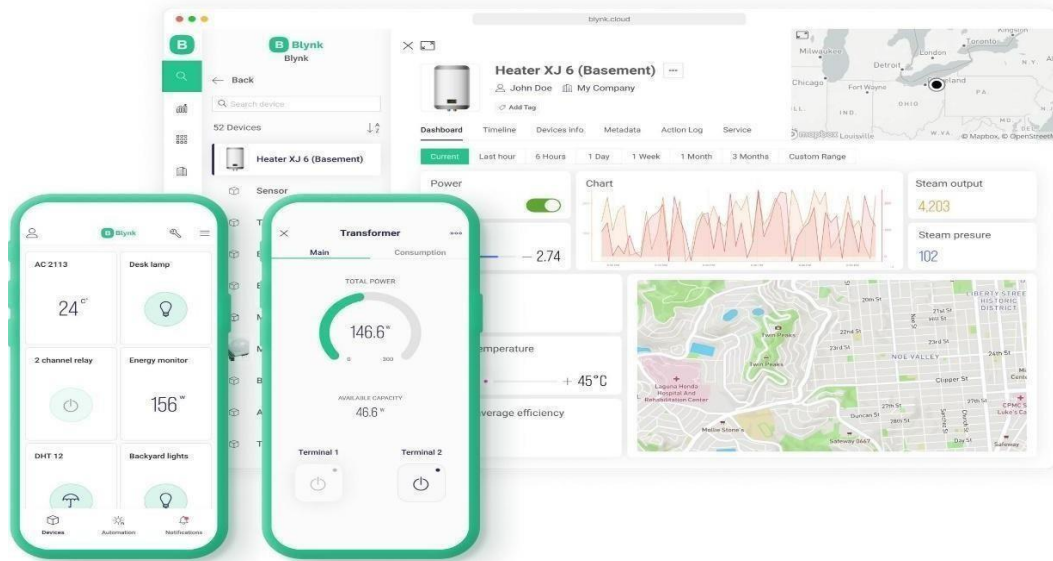
Overall, the Arduino IDE is a powerful software tool that provides a user- friendly interface for writing, compiling, and uploading Arduino sketches to the board. It is widely used by hobbyists, students, and professionals for prototyping and developing electronics Projects.

4.1.1 Programming Language

C programming language is a widely used programming language that is often used in embedded systems, including Arduino development. Arduino is an open- source platform that provides hardware and software tools for building and programming microcontroller-based projects. To use C programming language with Arduino, you would typically use the Arduino Integrated Development Environment (IDE), which is a software tool that provides a user-friendly interface for writing, compiling, and uploading C code to Arduino boards.

4.1.2 Blynk IoT

Blynk IoT is a cloud-based platform used for developing and managing Internet of Things (IoT)



applications. It enables users to remotely monitor and control connected devices through a mobile application and web dashboard. Blynk provides a simple and efficient way to exchange data between hardware devices and users over the internet. It supports various microcontrollers such as ESP32, ESP8266, and Arduino, making it widely used in IoT projects. The platform offers real-time data visualization, remote device management, notifications, and secure cloud connectivity. In IoT projects, Blynk helps users access sensor data, monitor system performance, and control devices from anywhere using an internet connection.

4.2 Functional Requirements

- ✓ The system must employ sensors to continuously monitor both environmental conditions and the health parameters of mining workers in real-time.
- ✓ The smart helmet should be capable of detecting abnormal conditions, such as hazardous gases or sudden falls, through the use of sensors like gas sensors and MEMS accelerometers.
- ✓ In case abnormal conditions are detected, the system should generate immediate alerts within the helmet to notify the mining workers of potential dangers.
- ✓ The system should have the capability to send notifications to supervisors or relevant personnel, providing them with crucial information about the detected abnormal conditions

4.3 Non-Functional Requirements

4.3.1 Usability

The smart helmet must consistently and accurately operate in diverse mining environments and conditions, ensuring that workers can rely on its functionality during critical situations.

4.3.2 Portability

The project's portability refers to its ability to send alerts without being restricted by the user's internet access. This ensures that alerts can be communicated seamlessly regardless of the worker's location within the mining site.

4.3.3 Speed

The smart helmet is designed for use in mining scenarios, specifically to enhance safety and rescue capabilities for workers during emergencies. Its usability is crucial for providing effective and timely responses to potential risks in the mining environment.

4.3.4 Real-time Responsiveness

The system must provide real-time alerts and data updates, facilitating swift responses to emergencies or changing conditions within the mining environment.

4.4 Performance

Advanced sensors, including MEMS, guarantee accurate tracking of environmental conditions, providing real-time data for immediate response and worker safety.

The integration of cutting-edge communication modules like GSM and GPS enhances the reliability and responsiveness of the alert system, ensuring seamless connectivity in diverse mining environments.

A robust Wi-Fi module is employed for efficient data retention, enabling future references and data-driven analyses for continuous improvement of safety measures in mining operations.

The addition of an APR33A3 voice module ensures precise and clear voice alerts, specifying the exact cause during alerts. This feature enhances understanding and enables quick, appropriate responses to identified issues, fostering a safer working environment.

4.5 Methodology

To implement this project Iterative model is used. It involves continuous cycle of Planning, Analysis, Implementation and Evaluation, Testing, Deployment, Review, etc., Each cycle produces a segment of development that forms the basis for the next cycle of iterative development.

The iterative model is a software development life cycle (SDLC) approach in which initial development work is carried out based on well-stated basic requirements, and successive enhancements are added to this base piece of software through iterations until the final system is built.

4.5.1 Advantages

- It is easily acceptable to ever-changing needs of the project.
- Testing and debugging during smaller iteration are easy.
- A parallel development can plan.
- Risks are identified and resolved during iteration.
- Limited time spent on documentation and extra time on designing

4.6 CostEstimation

S. No	Component	Cost (Rs)	No of pieces required	Total Cost (Rs)
1	Helmet	350	1	350
2	Battery	200	1	200
3	GPS Module	650	1	650
4	Microcontroller ESP32	650	1	650
5	DHT 11 Sensor	275	1	275
6	Gas Sensor	270	1	270
7	MPU6050 Sensor	150	1	150

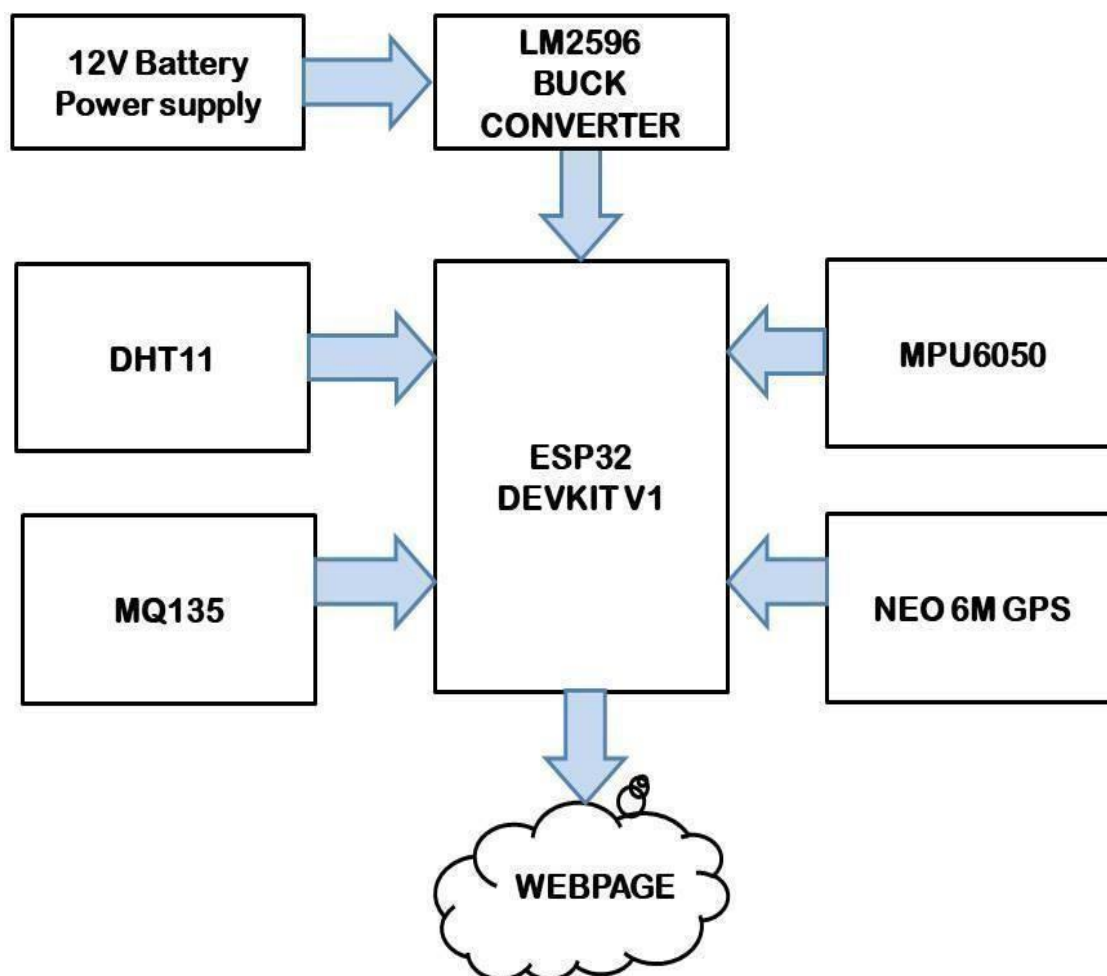
4.7 Time Estimation

S. No	Activity	Duration
1	Domain & Title	1 Week
2	Literature Survey	1 Week
3	Requirements Specification	1 Week
4	Planning	1 Week
5	Design	1 Week
6	Gathering Requirements	1 Week
7	Software Development	1 Week
8	Hardware Development	2 Weeks
9	Testing	1 Week
10	Implementation	1 Week

CHAPTER 4

DESIGN

4.1 System Architecture



The block diagram illustrates a sensor-based embedded system built around the ESP32 Devkit V1 microcontroller. The system is powered by a 12V battery, which is stepped down to a suitable voltage for the ESP32 and other components using an LM2596 buck converter. The ESP32 serves as the central processing unit, interfacing with multiple sensors and modules. A DHT11 sensor is connected to measure temperature and humidity, while an MQ135 gas **sensor** detects air quality and harmful gases. The MPU6050 module provides motion sensing through its accelerometer and gyroscope, and the NEO-6M GPS module supplies location data. The ESP32 processes the collected data and transmits it to a webpage for remote monitoring, enabling real-time access to environmental and positional information.

Flow Chart

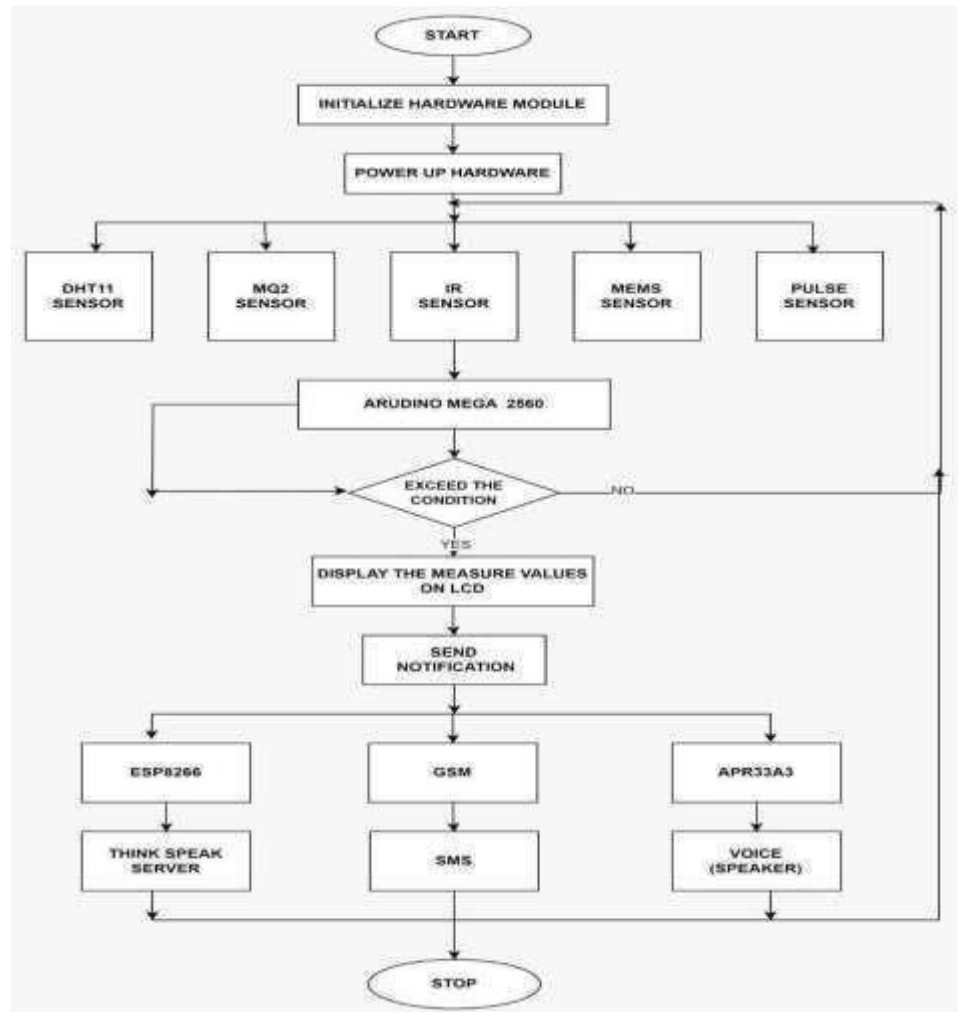


Fig. Flow Chart

The flowchart provides a comprehensive insight into the operational intricacies of the smart helmet system designed for mining safety. The journey commences with the initiation phase, where all hardware components, including the pivotal Arduino Mega microcontroller, an array of sensors, and versatile communication modules, are primed for action. This initialization sets the stage for a dynamic and interconnected system that aims to ensure the well-being of mining workers in challenging environments.

As the hardware powers up, the flowchart elegantly bifurcates into dedicated branches for each crucial sensor embedded in the smart helmet. These sensors, namely the DHT11 for temperature and humidity, MO2 for gas detection, IR for helmet usage, MEMS for fall detection, and the Pulse sensor for heart rate monitoring, collectively form a robust sensory network. Each sensor plays a vital role in assessing specific aspects of

the mining environment and the wearer's physiological parameters.

The system then meticulously evaluates readings from each sensor, gauging whether they breach predefined threshold levels indicative of potential risks. Should a sensor reading exceed the established safety thresholds, the system follows a cascading sequence of actions. It includes the real-time display of the measured value on the helmet's LCD screen, providing immediate feedback to the wearer. Simultaneously, the system initiates notifications, offering a versatile approach through options like GSM for SMS alerts and ESP8266 Wi-Fi for cloud-based communication.

Conversely, if sensor readings fall within acceptable limits, the system seamlessly proceeds through the evaluation without triggering alarms. This cyclical and continuous process ensures perpetual monitoring of the mining environment and the miners' well-being. By seamlessly integrating cutting-edge technology, the smart helmet system stands as a testament to innovation in occupational safety, providing real-time insights and actionable alerts in potentially hazardous scenarios.

CHAPTER 6

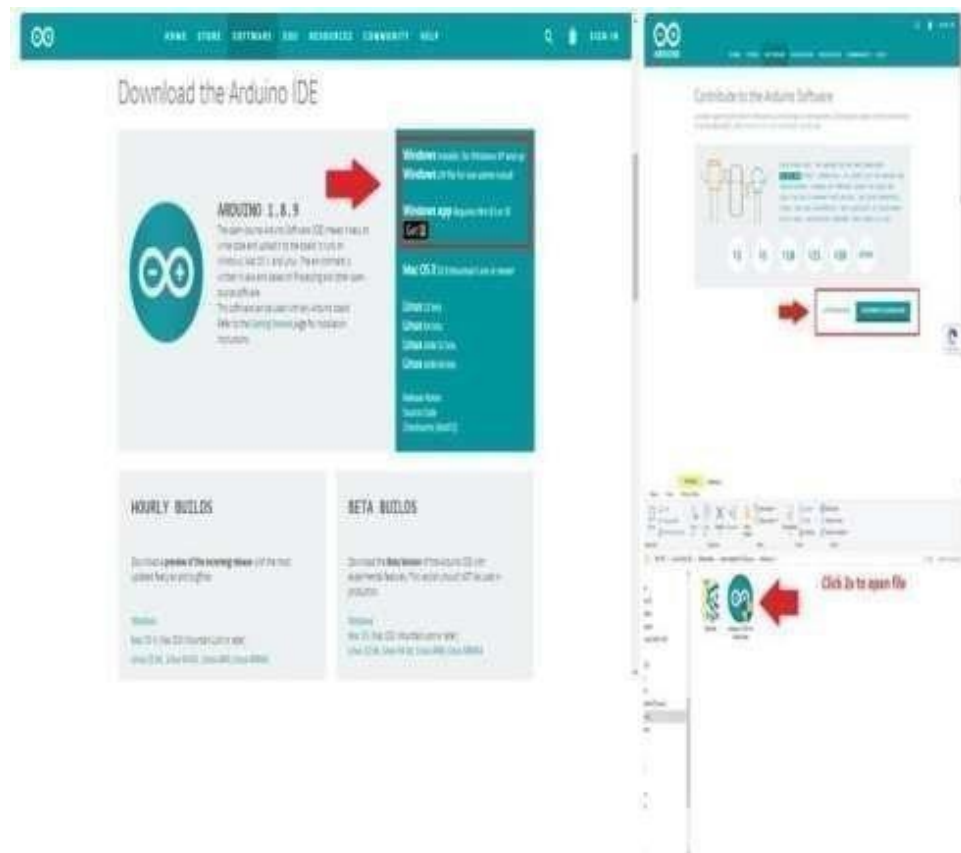
IMPLEMENTATION

Software Implementation

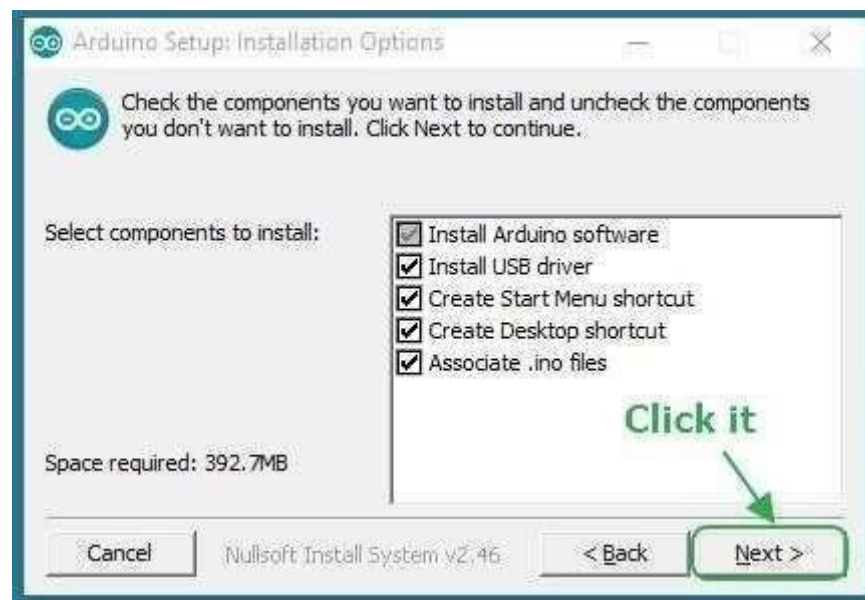
Installation of Arduino UNO IDE

To install Arduino UNO IDE on your Windows PC, follow the next instructions:

STEP 1: Download file Arduino IDE



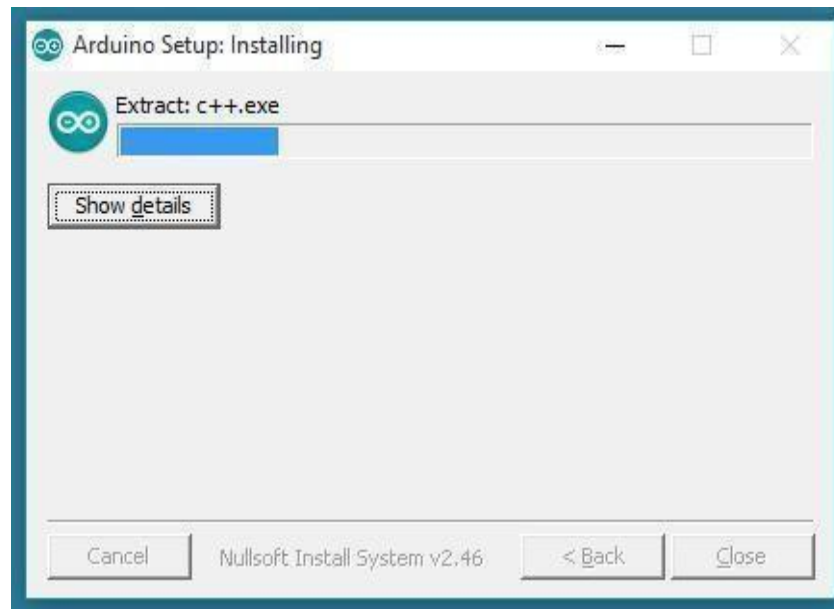
STEP 2: Installation option



STEP 3: Installation folder



STEP 4: Installing process



STEP 5: Get an Arduino Mega and USB cable

In this instructional exercise, you're utilizing an Uno R3. You additionally require a standard USB link (A fitting to B plug): the kind you would associate with a USB printer



Fig : USB cable and Arduina Mega board

STEP 6: Connect the board

The USB association with the PC is important to program The USB association with the PC is important to program the board and not simply to control it up. The Uno and Mega consequently draw control from either the USB or an outside power supply. Associate the board to your PC utilizing the USB link.

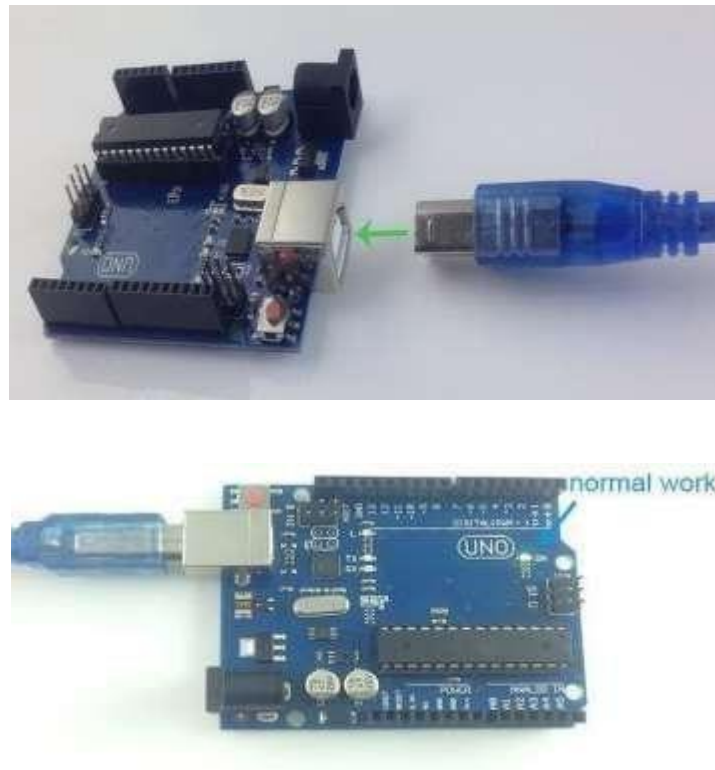


Fig : Represents how to connect USB cable with Mega board

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STEP 7: Select your board

You'll need to select the entry in the Tools > Board menu that corresponds to your Arduino board.

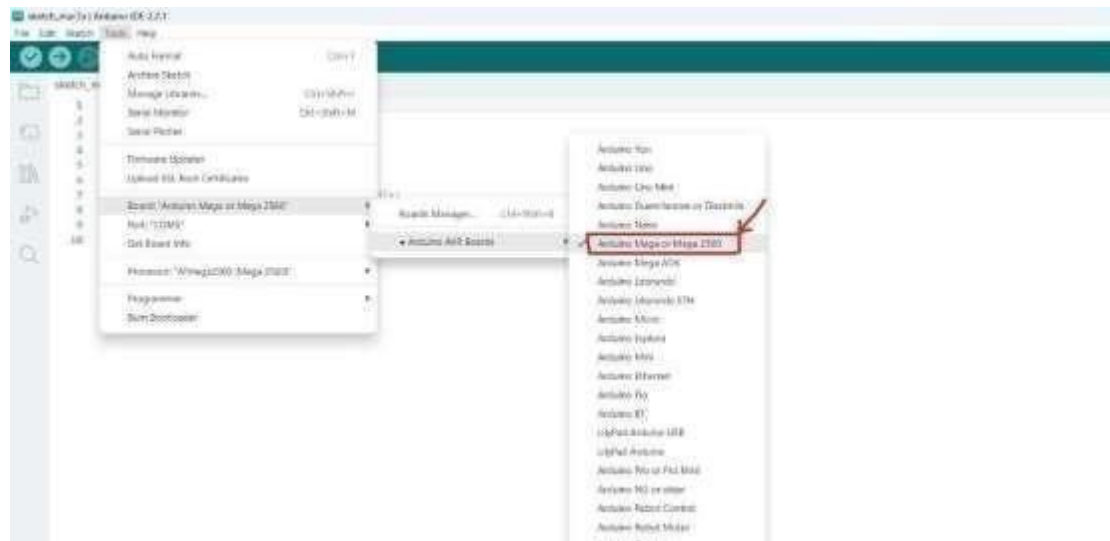


Fig : Path to Select Board

STEP 8: Add library file: Sketch>Include Library>Add.ZIP Library

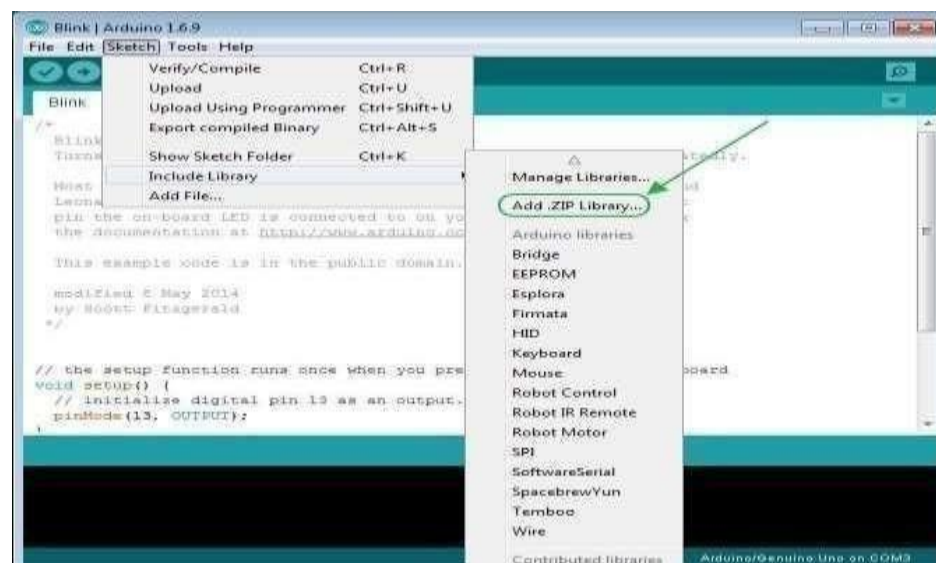


Fig : Path to Add Zip Library

STEP 9: Select your library file compression package on the demo code file, as follows

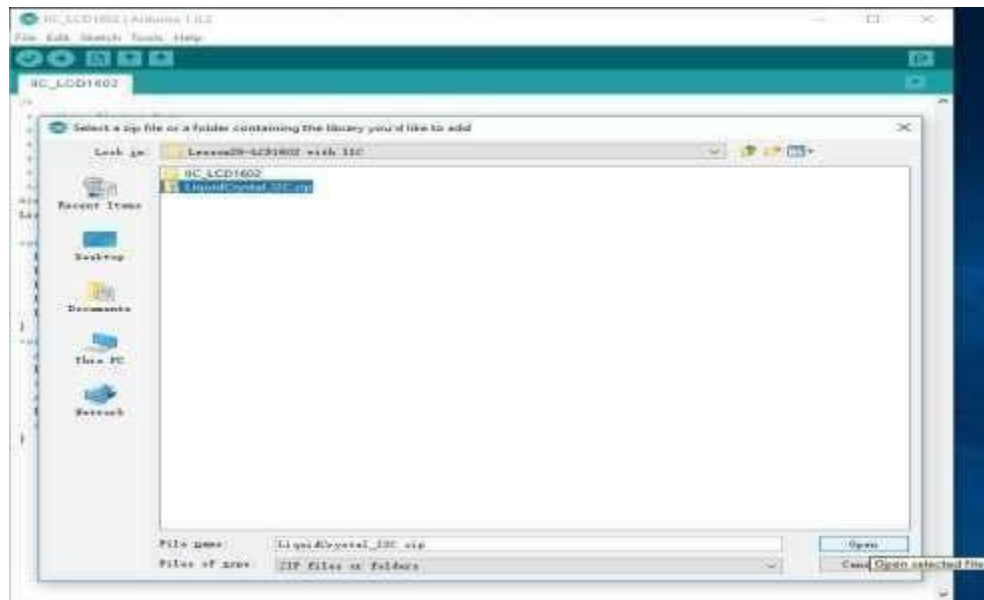


Fig : Selecting a Zip File

STEP 10: Select your serial port

Select the serial gadget of the board from the Tools | Serial Port menu. This is probably going to be COM3 or higher (COM1 and COM2 are generally held for equipment serial ports). To discover, you can detach your board and re-open the menu; the passage that vanishes ought to be the Arduino board. Reconnect the board

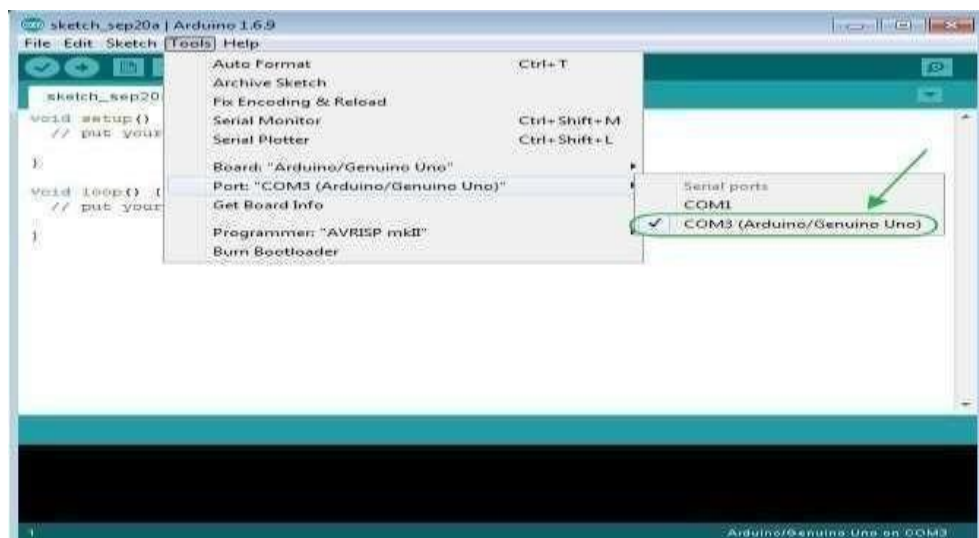


Fig : Path to Select Serial Port

6.1 Project Source Code

```
#include <WiFi.h> #include
<WebServer.h> #include
<Wire.h> #include
<TinyGPS++.h>
#include <HardwareSerial.h>
#include <DHT.h>

// ===== WiFi AP =====
const char* ssid =
"SmartHelmet"; const char*
password = "12345678";

WebServer server(80);

// ===== GPS =====
TinyGPSPlusgps;
HardwareSerial
gpsSerial(2);

// ===== DHT =====
#define DHTPIN 4
#define DHTTYPE
DHT11
DHT dht(DHTPIN, DHTTYPE);

// ===== MQ4 =====
#define MQ4_DO 34

// ===== MPU6050 =====
const int MPU_ADDR =
0x68; int16_t AcX;

// ===== Variables =====
float temperature
= 0; float
humidity = 0;
String gasStatus
= "Normal";
String fallStatus
= "OK";

// ===== HTML PAGE =====
String webpage = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
```

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```
<title>Smart Helmet Dashboard</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
body { font-family: Arial; text-align: center; background: #111; color: white; }
.card { background: #222; padding: 20px; margin: 10px; border-radius:
10px; } h1 { color: cyan; }
.alert { color: red; font-weight: bold; }
.ok { color: lime; }
</style>

</head>

<body>
<h1>Smart Helmet Dashboard</h1>

  <div class="card">
    <h2>Temperature</h2>
    <p id="temp">--</p>
  </div>

  <div class="card">
    <h2>Humidity</h2>
    <p id="hum">--</p>
  </div>

  <div class="card">
    <h2>Gas Status</h2>
    <p id="gas">--</p>
  </div>

  <div class="card">
    <h2>Fall Status</h2>
    <p id="fall">--</p>
  </div>

  <div class="card">
    <h2>Location</h2>
    <p id="gps">--</p>
  </div>

  <script> setInterval(()
=> { fetch("/data")
.then(response => response.json())
  .then(data => {

document.getElementById("gas").innerHTML = data.gas;
document.getElementById("fall").innerHTML = data.fall;
document.getElementById("gps").innerHTML = data.lat + ", " +
data.lng;
```

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```
document.getElementById("gas").className = (data.gas=="GAS DETECTED")
? "alert" : "ok";
document.getElementById("fall").className = (data.fall=="FALL DETECTED")
? "alert" : "ok";
}, });
2000);
</script>

</body>
</html>
)rawliteral";

// ===== HANDLERS =====
void handleRoot() {
server.send(200, "text/html", webpage);
}

void handleData()
{ String json = "{";
json += "\"temp\":" + String(temperature) +
","; json += "\"hum\":" + String(humidity)
+ ","; json += "\"gas\":" + gasStatus +
"\","; json += "\"fall\":" + fallStatus +
"\","; json += "\"lat\":" + latitude
+ "\","; json += "\"lng\":" +
longitude + "\""; json += "}";

server.send(200, "application/json", json);
}

// ===== SETUP =====
void setup()
{ Serial.begin(115200);

  WiFi.softAP(ssid,
password);
Serial.println(WiFi.softAPIP());

  server.on("/", handleRoot); server.on("/data",
handleData); server.begin();

  dht.begin();
pinMode(MQ4_DO,
INPUT);

  Wire.begin(21, 22);
Wire.beginTransmission(MPU_ADDR);
Wire.write(0x6B);
Wire.write(0);
Wire.endTransmission(true);
gpsSerial.begin(9600,
```

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```
}

// ===== LOOP =====

void loop() {

  // GPS
  while (gpsSerial.available())
  { gps.encode(gpsSerial.read());
  }

  if (gps.location.isUpdated()) {
    latitude = String(gps.location.lat(), 6);
    longitude = String(gps.location.lng(),
    6);

    }
    // else → default location remains

  // DHT
  temperature
  =
  dht.readTemperature();
  humidity =
  dht.readHumidity();

  // MQ4
  if (digitalRead(MQ4_DO) == LOW)
  { gasStatus = "GAS DETECTED";
  } else {
    gasStatus = "Normal";}

  // MPU6050
  Wire.beginTransaction(MPU_ADD
  R); Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(MPU_ADDR, 2,
  true);

  AcX = Wire.read() << 8 |

  Wire.read(); if (abs(AcX) >

  10000) {
    fallStatus = "FALL DETECTED";
  } else {
    fallStatus = "OK";
  }

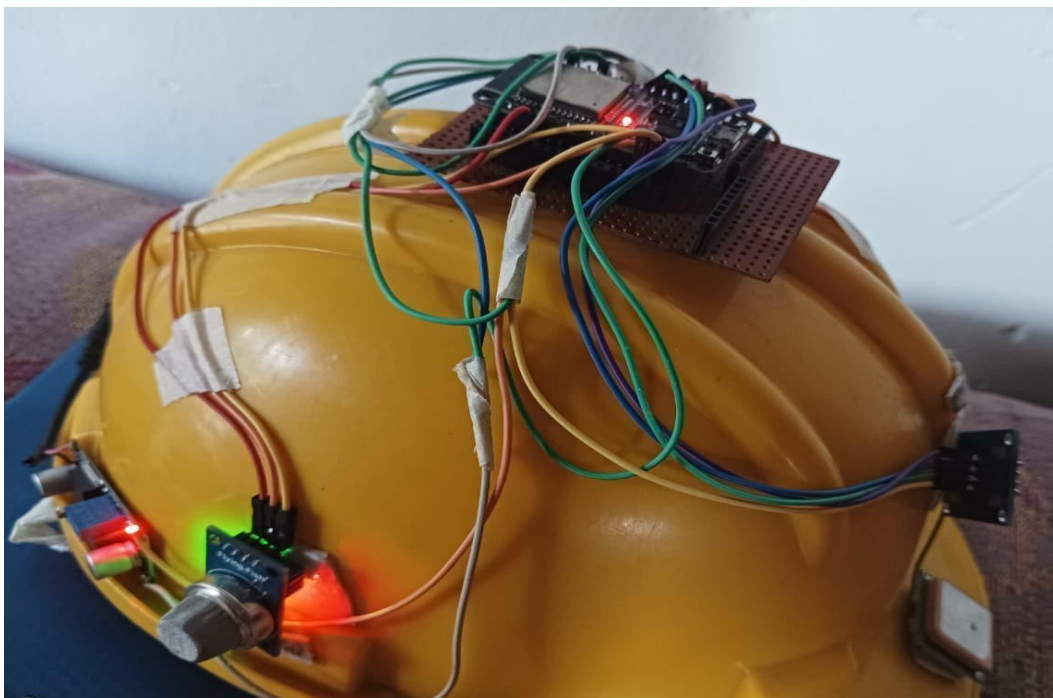
  server.handleClient();
}
```

CHAPTER 7

RESULTS

7.1 Results

In the Smart Helmet project, the core functionality revolves around the integration of various sensors and components into a wearable helmet designed to enhance safety and monitoring for mining workers. The system incorporates specialized sensors such as gas sensors, temperature and humidity sensors, accelerometers, gyroscopes, infrared sensors, and heart rate sensors. These sensors work collectively to monitor environmental conditions and the well-being of the wearer in real-time. For instance, gas sensors detect hazardous gases, temperature and humidity sensors monitor environmental parameters, while accelerometers and gyroscopes track worker movements and detect falls. The data collected by these sensors is processed by a microcontroller embedded within the helmet. Upon detecting abnormal conditions or emergencies, the microcontroller triggers alerts and notifications, ensuring timely response and assistance for the wearer. Additionally, the helmet features a voice interface module for delivering alerts and instructions, further enhancing communication and safety in mining environments. Overall, the Smart Helmet project aims to provide comprehensive safety measures and real-time monitoring to safeguard mining workers' well-being.



CHAPTER 8

SOFTWARE TESTING

Types Of Testing

The testing of the Smart Helmet project will be conducted systematically to ensure its functionality and reliability. It will be divided into two main stages: software testing and hardware testing. Each stage will involve specific features to be tested, testing tools, and procedures.

Features to be Tested

The following features of the Smart Helmet project will be tested:

Environmental Sensors:

Gas sensor: Ensure it detects hazardous gases accurately.

Temperature and humidity sensor: Verify it monitors environmental conditions effectively.

Worker Condition Monitoring:

Heart rate sensor: Validate its accuracy in tracking the worker's heart rate.

Infrared sensor: Ensure it confirms continuous helmet usage.

MEMS sensor: Verify fall detection capabilities.

Alerting System:

APR33A3 voice module: Confirm it provides clear and specific alerts.

GSM module: Ensure timely transmission of alert messages.

GPS module: Validate accurate location tracking during emergencies.

Testing Tools and Environment

For testing the Smart Helmet project, the following tools and environment will be utilized:

Arduino IDE: To test the software functionality and code execution.

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Power supply: To provide power to the helmet system during testing.

Testing environment: A simulated mining environment or controlled laboratory setting to replicate real-world conditions.

Test Cases

In this section, we will define test cases to validate the functionality of the system:

Inputs

The inputs for testing the Smart Helmet project include:

Power supply: Ensure sufficient power is provided to all components.

Simulated environmental conditions: Test various scenarios to validate sensor responses.

Worker conditions: Mimic different worker scenarios to verify sensor accuracy.

Expected Output

The expected outputs of the Smart Helmet project include:

Accurate detection and monitoring of environmental conditions.

Real-time tracking and monitoring of worker health parameters.

Timely and clear alert messages during hazardous events.

Testing Procedure

The testing procedure for the Smart Helmet project involves the following steps:

Software Testing:

Upload the code to the Arduino or microcontroller.

Test various scenarios to test sensor functionality.

Verify the alerting system by triggering different events.

Hardware Testing:

Assemble the helmet with all components properly connected.

Field Testing:

Deploy the Smart Helmet in a real mining environment.

Monitor its performance in detecting and responding to actual hazards. Collect feedback from miners for further improvements.

By following this testing approach, the Smart Helmet project can be thoroughly evaluated for its effectiveness in enhancing miner safety and well-being.

CHAPTER 9

CONCLUSION

In conclusion, the integration of IoT-based smart helmets in the mining industry marks a significant advancement in ensuring the safety and well-being of miners. The existing system, relying on traditional helmets, demonstrates certain limitations, particularly in maintaining environmental awareness, fall detection, and specifying the exact cause during alerts. The proposed smart helmet system addresses these gaps by integrating various sensors, including a gas sensor and temperature/humidity sensor for environmental monitoring, ensuring protection against hazardous conditions. Additionally, the helmet utilizes MEMS sensors, an IR sensor, and a heart rate sensor to monitor workers' conditions. Moreover, the system is enhanced by an efficient alerting system, incorporating a voice interface module (APR33A3) and robust communication system by GSM and GPS modules. To facilitate data retention for future references and predictive analyses, a Wi-Fi module is employed to store information on ThingSpeak. This holistic system represents an advanced solution that elevates workplace safety through real-time monitoring, precise alerts, and insights derived from data analysis. Beyond safety enhancement, this comprehensive solution also contributes to improved communication, emergency response, and overall operational efficiency within mining endeavors.

CHAPTER 10

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ABSTRACT
Mining is recognized as one of the most hazardous occupations globally. The death rate of mining workers at mining sites is increasing day by day. Considering this, we cannot simply avoid mining because it plays a vital role. Instead, an alternative solution is to implement safety measures and continue operations in mines. To ensure the well-being of workers

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Instead of relying on traditional loud alarms, the helmet communicates crucial information to the miner through a voice interface, offering real-time updates and guidance during emergency situations.